OpCode	6-bit coding	Instruction	ALUOp	3-bit Coding
0	000000	ADD	ADD	000
1	000001	SUB	SUB	100
2	000010	AND	AND	001
3	000011	OR	OR	010
4	000100	XOR	XOR	011
5	000101	NOR	NOR	101
6	000110	SLL	SLL	110
7	000111	SRL	SRL	111
8	001000	CLR	X	X
9	001001	JR	X	X
10	001010	ADDI	ADD	000
11	001011	ANDI	AND	001
12	001100	ORI	OR	010
13	001101	LW	ADD	000
14	001110	SW	ADD	000
15	001111	SWAP	ADD	000
16	010000	BEQ	XOR	011
17	010001	BNE	XOR	011
18	010010	J	X	X
19	010011	JAL	X	X

OpCode	6-bit coding	Instruction	RegDst	LinkSel	ExtOp	RegW	ALUSrc	MemRd	MemWr	Wbdata
0	000000	ADD	1	0	X	1	0	0	0	00
1	000001	SUB	1	0	X	1	0	0	0	00
2	000010	AND	1	0	X	1	0	0	0	00
3	000011	OR	1	0	X	1	0	0	0	00
4	000100	XOR	1	0	X	1	0	0	0	00
5	000101	NOR	1	0	X	1	0	0	0	00
6	000110	SLL	1	0	X	1	0	0	0	00
7	000111	SRL	1	0	X	1	0	0	0	00
8	001000	CLR	1	0	X	1	X	0	0	11
9	001001	JR	X	X	X	0	X	0	0	X
10	001010	ADDI	0	0	1	1	1	0	0	00
11	001011	ANDI	0	0	0	1	1	0	0	00
12	001100	ORI	0	0	0	1	1	0	0	00
13	001101	LW	0	0	1	1	1	1	0	01
14	001110	SW	X	X	1	0	1	0	1	X
15	001111	SWAP	0	0	1	1	1	1	1	01
16	010000	BEQ	X	X	1	0	0	0	0	X
17	010001	BNE	X	X	1	0	0	0	0	X
18	010010	J	X	X	X	0	X	0	0	X
19	010011	JAL	X	1	X	1	X	0	0	10

OpCode	6-bit coding	Instruction	Z flag	PCSrc
0	000000	ADD	X	00
1	000001	SUB	X	00
2	000010	AND	X	00
3	000011	OR	X	00
4	000100	XOR	X	00
5	000101	NOR	X	00
6	000110	SLL	X	00
7	000111	SRL	X	00
8	001000	CLR	X	00
9	001001	JR	X	11
10	001010	ADDI	X	00
11	001011	ANDI	X	00
12	001100	ORI	X	00
13	001101	LW	X	00
14	001110	SW	X	00
15	001111	SWAP	X	00
16	010000	BEQ	0	00
16	010000	BEQ	1	10
17	010001	BNE	0	10
17	010001	BNE	1	00
18	010010	J	X	01
19	010011	JAL	X	01

Suppose Opcode = A

Control Ssignal	Logic Equation
ALUOp2	A4′A3′A1′A0+A1′A2A3
ALUOp1	A4+A3′A2A0′+(A3′+A2)(A1⊕A0)′
ALUOp0	A4+A3′A2′A1A0′+A3′A2A1′+A0(A3⊕A2)
RegDst	A4′(A3′+A2′A1′)
LinkSel	A4
ExtOp	A4+A2A1+(A1⊕A0)
RegW	A4′A3′+A4′A1′A0′+(A2⊕A1)+A1A0
ALUSrc	A3
MemRd	A3A2A0
MemWr	A3A2A1
Wbdata1	A4+A3A2′A1′
Wbdata0	A3(A2⊕A1)′+A1′A0
PCSrc1	PCSrc1=A3A2′A1′A0+A4A1′(Z⊕A0)
PCSrc0	A3A2′A1′A0+A4A1

For Pipeline

Control Ssignal	Logic Equation
ForwardA1	RegWr3·NZ(Rd3)·EQ(Rd3,Rs2)
ForwardA0	RegWr4·NZ(Rd4)·EQ(Rd4,Rs2)·ForwardA1'
ForwardB1	RegWr3·NZ(Rd3)·EQ(Rd3,Rt2)
ForwardB0	RegWr4·NZ(Rd4)·EQ(Rd4,Rt2)·ForwardB1'
Stall	MemRd2·(EQ(Rt2,Rs1)+EQ(Rt2,Rt1))
IF/ID Flush	BranchTaken+J+JR
ID/EX Flush	Stall+BranchTaken
Branch Taken	BEQ·EQ(BusA,BusB)+BNE·EQ(BusA,BusB)'
PCWrite	Stall'
IFIDWrite	Stall'